Year/period	Drone model/system	Developer/country	Key technological features	Prima applic
			+ strike), cost-effective design	

4.3 Principles and Components of Third-generation Drones

The third generation of unmanned aerial vehicles (UAVs) represents a convergence of advanced avionics, networked communications, multi-sensor payloads, and persistent autonomy. Where earlier generations extended range and added basic automation, third-generation systems integrate real-time decision support, redundant flight management, and global connectivity to perform complex, long-duration missions with high reliability. This section outlines the operating principles that make those capabilities possible and describes the principal hardware and software subsystems that constitute a contemporary third-generation drone.

Figure 4.1 illustrates the key structural and functional components that define the third generation of unmanned aerial vehicles (UAVs). The labeled diagram highlights the integration of critical systems such as the multi-sensor payload, satellite communication antenna, redundant flight control systems, autonomous flight software, onboard processing unit, propulsion system, and ground control station. Together, these components represent the synergy between hardware and software that enables high-level autonomy, endurance, and real-time data exchange. The multi-sensor payload ensures all-weather intelligence gathering, while the SATCOM antenna facilitates beyond-lineof-sight operations. The autonomous flight software and redundant flight control units provide mission reliability and fault tolerance. The onboard processing unit handles sensor fusion and decision-making tasks, while the propulsion system supports efficient, long-